



Glasses and their crystallization in the $(1 - x)\text{KNbO}_3 \cdot x\text{SiO}_2$ system at low glass-forming oxide contents, $0 \leq x \leq 0.35$

V.N. Sigaev^{a,*}, V.S. Ryzhenkov^a, S.V. Lotarev^a, N.V. Golubev^a, S. Yu. Stefanovich^b, Akira Okada^c

^aD.I. Mendeleev University of Chemical Technology of Russia, Optical Glass Center, Miusskaya sq. 9, 125047 Moscow, Russia

^bL.Ya. Karpov Institute of Physical Chemistry, Vorontsovo pole 10, 103064 Moscow, Russia

^cNissan Research Center, Nissan Motor Co. Ltd., Natsushima-cho, Yokosuka 237-8523, Japan

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ABSTRACT

Glasses based on the $\text{K}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ system have been a subject of numerous studies thanks to promising electro-optical and non-linear optical properties. Most of the studies of Nb-containing glasses deal with compositions including not less than 40 mol% of SiO_2 or B_2O_3 . In the present work, we studied glass-forming and crystallization properties of $(1 - x)\text{KNbO}_3 \cdot x\text{SiO}_2$ glasses with x varying from 0 to 35 mol% by melt quenching between steel plates or by exposure to air or nitrogen flow. Thin transparent glassy plates can be obtained by quenching via pressing between metal plates for glasses with $x \geq 0.12$, and transparent glass-ceramics allowing a remarkable second harmonic generation (SHG) response can be formed by heat-treatment near the first exothermic differential thermal analysis (DTA) peak over a wide range of glass compositions. It is shown that in these glasses, the crystalline $\text{K}_4\text{Nb}_6\text{O}_{17}$ phase and a small amount of ferroelectric KNbO_3 , along with ceolite-like, cristobalite and α -quartz SiO_2 phases precipitate at low temperatures in the range between the glass transition point, T_g , and the first exothermic peak on DTA curve, whereas ferroelectric KNbO_3 precipitates at temperatures above 800 °C. Ferroelectric crystals which precipitate in the glass bulk are shown to produce high SHG activity.

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1. Introduction

Glasses in $\text{Me}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ (B_2O_3 , P_2O_5 , GeO_2) systems with $\text{Me} = \text{Li}$, Na , K have been a subject of numerous studies. These glasses allow the production of electro-optical [1–3] and non-linear optical [4–7] activity. Zhilin et al. [8,9] obtained transparent glass-ceramics and demonstrated the creation of gradient optical materials in $\text{Na}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ system glasses. Polarization of Nb-containing glasses in dc-electric field at elevated temperatures (thermal poling) can produce a highly efficient non-linear optical layer whose second-order optical non-linearity coefficient d_{33} reaches 5 pm/V value [10] and hence is already comparable with that of KDP and other popular non-linear optical crystals [11,12].

An important feature which determines the properties of Nb-containing glasses is the high polarizability and hyperpolarizability of the Nb atoms and also the large anisotropy of $[\text{NbO}_5]$ and $[\text{NbO}_6]$ polyhedra occurring with a large (up to 0.55 Å) dispersion of Nb–O bond lengths [13,14]. LiNbO_3 crystals can precipitate in $\text{Li}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ (B_2O_3 , GeO_2) glasses. In initial stages of crystallization, these crystals form either transparent surface lay-

ers [4,15,16] or bulk nanostructures [4,16,17]. Nanocrystals possessing tetragonal tungsten bronze structure precipitate in $\text{Na}_2\text{B}_4\text{O}_7-\text{Nb}_2\text{O}_5$ glasses [11]. Potassium-containing glasses of the family being discussed exhibit the crystallization of ferroelectric potassium niobate (KNbO_3) whose efficiency in light frequency conversion (e.g. second harmonic generation – SHG) is even higher than that of LiNbO_3 .

Unfortunately, KNbO_3 has never been obtained from the glass as the first crystallizing phase before. Tanaka et al. [18] showed that in $40\text{K}_2\text{O}-25\text{Nb}_2\text{O}_5-35\text{SiO}_2$ glass KNbO_3 precipitates at temperatures above 690 °C when the glass is already non-transparent. Recently KNbO_3 was reported to precipitate as the first crystalline phase during heat-treatment near the first exothermic DTA peak in $40\text{K}_2\text{O}-20\text{Nb}_2\text{O}_5-40\text{B}_2\text{O}_3$ glass [19] but this phase was found to be cubic (JCPDS card #32-0821). Glass-ceramics based on this phase are assumed to reveal no second-order optical non-linearity. It is also established that Nd:YAG laser irradiation of $40\text{K}_2\text{O}-40\text{Nb}_2\text{O}_5-20\text{SiO}_2$ glass doped by small amount of NiO results in surface crystallization of the metastable KNbO_3 phase (JCPDS card #32-0821) but ferroelectric KNbO_3 is not observed [20]. Moreover, in a recent study, homogeneous $23\text{K}_2\text{O}-27\text{Nb}_2\text{O}_5-50\text{SiO}_2$ glass was shown to possess high second-order optical non-linearity up to $\chi^{(2)} = 3.8 \text{ pm/V}$ after being thermally poled [21].

* Corresponding author.

E-mail address: vlad.sigaev@gmail.com (V.N. Sigaev).

Models suggested to explain high second-order optical non-linearity of thermally poled Nb-containing glasses are usually based on ideas involving inner electric field induced by partial migration of monovalent cations to the cathode and on the high hyperpolarizability of distorted niobium–oxygen polyhedra [18,19,22]. Therefore, one can assume that poling efficiency rises along with an increase of their content. According to [23,24] glass-forming materials in $\text{Me}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ (B_2O_3 , P_2O_5 , GeO_2) contain more than 20 mol% of glass-forming oxide but most of studies of Nb-containing glasses deal with compositions including not less than 40 mol% of SiO_2 or B_2O_3 .

It is quite possible that glasses in the abovementioned Nb_2O_5 -based systems can be obtained even at lower glass-forming oxide concentration by more rapid quenching of melts. Still we know nothing about the glass-forming process and crystallization of such glasses. Glass-former content minimization while retaining acceptable manufacturing conditions is extremely interesting both for clarification of the mechanisms producing optical non-linearity in glasses and also for developing technically promising materials.

The present study concerns glass-forming and crystallization of $\text{K}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ (KNS) glasses with $\text{K}_2\text{O}/\text{Nb}_2\text{O}_5 = 1$ and less than 40 mol% SiO_2 , obtained either by melt quenching between steel plates or by exposure to air or nitrogen flow at high pressure. High second-order optical non-linearity which can be obtained via thermal poling and preconditions for nanocrystallization of large amount of ferroelectric KNbO_3 make these glasses an object of noticeable scientific interest.

2. Experimental

In the present study efforts were made to obtain glass plates or amorphous particles of $(1-x)\text{KNbO}_3 \cdot x\text{SiO}_2$ compositions ($x = 0.35, 0.3, 0.25, 0.2, 0.15, 0.12, 0.1, 0.075, 0.05, 0.03, 0.015, 0$) named below as KNS- x with x substituted by corresponding SiO_2 percentage. Their positions are shown in the ternary diagram in Fig. 1. Glass-forming areas revealed in papers [23–25] for various melt cooling conditions are also represented: pressing of the melt on a metal plate [23] (melting in quartz crucible) and cooling of several grams of the melt in a platinum crucible on air [24]. Fig. 1 shows that so

far KNS glasses in the area of $\text{K}_2\text{O}/\text{Nb}_2\text{O}_5 = 1$ were obtained for SiO_2 content approximately down to 25 mol%.

Ultra pure K_2CO_3 (99.8%), Nb_2O_5 (99.997%), and amorphous SiO_2 (99.9%) compounds were used as initial reagents. A batch of 20 g of glass was prepared. The abovementioned glasses were melted in platinum crucibles for 20 min at temperatures from 1250 to 1400 °C depending on the composition. Glasses were obtained by pouring the melt onto a steel plate followed by pressing the melt with another steel plate. Glass samples were prepared as plates 0.5–1.5 mm thick. When it was not possible to obtain transparent samples by melt pressing, a falling melt stream was exposed to air or nitrogen flow under the pressure of 7–25 MPa resulting in small droplets, fibers and flakes. In all cases, the amorphous or crystalline state of as-quenched or heat-treated samples was revealed by X-ray diffraction (XRD) measurements (DRON-3M diffractometer, $\text{Cu K}\alpha$ radiation, nickel filter). All the amorphous samples obtained were investigated by differential thermal analysis (DTA) in air at heating rate of 10 K min^{-1} in 20–1000 °C temperature range using Al_2O_3 powder as a reference sample. Precipitation of non-centrosymmetric phases in glasses was revealed by the SHG-technique [26]. A Minilite-I pulsed solid-state laser operating in Q-switching mode at repetition frequency of 6 Hz (1.064 μm wavelength; pulse power, approximately 0.1 MW; pulse width, 3 ns) was used as a radiation source. The reported SHG signal ($I_{2\omega}$) indicates the ratio of second harmonic radiation intensity of the tested sample to that of the reference α -quartz powder standard (particle size: 3–5 μm). Scanning electronic microscopy (SEM) was carried out using JSM 7500 FA high resolution scanning electron microscope (JEOL, Japan) in a secondary electrons mode.

3. Results

Glass plates with no signs of crystallization were obtained by melt pressing for compositions in the range of x from 0.35 to 0.12. For $x < 0.12$ they were partially or completely crystallized. Melt exposure to high velocity air flow was used for the compositions with lower SiO_2 content. The amorphous droplets, fibers and flakes have been obtained for all investigated glasses down to KNbO_3 ($x = 0$). Amorphous structure of the samples obtained was

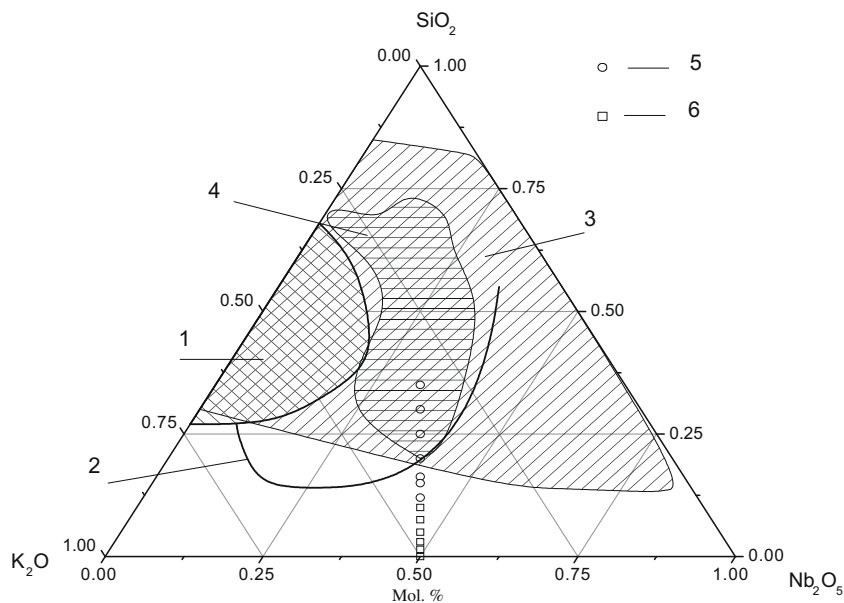


Fig. 1. Glass-forming area for $\text{K}_2\text{O}-\text{Nb}_2\text{O}_5-\text{SiO}_2$ system for various cooling conditions of cooling: 1 – glasses with low chemical resistance [23]; 2 – glasses obtained by pressing the melt on a metal plate [23] (melting in the quartz crucible); 3 – glasses obtained by cooling several grams of melt in the platinum crucible on air [24]; 4 – compositions studied in [25]. Glasses of compositions studied in the present work with $\text{K}_2\text{O}/\text{Nb}_2\text{O}_5 = 1$ obtained by pressing (5) and blowing of melt (6).

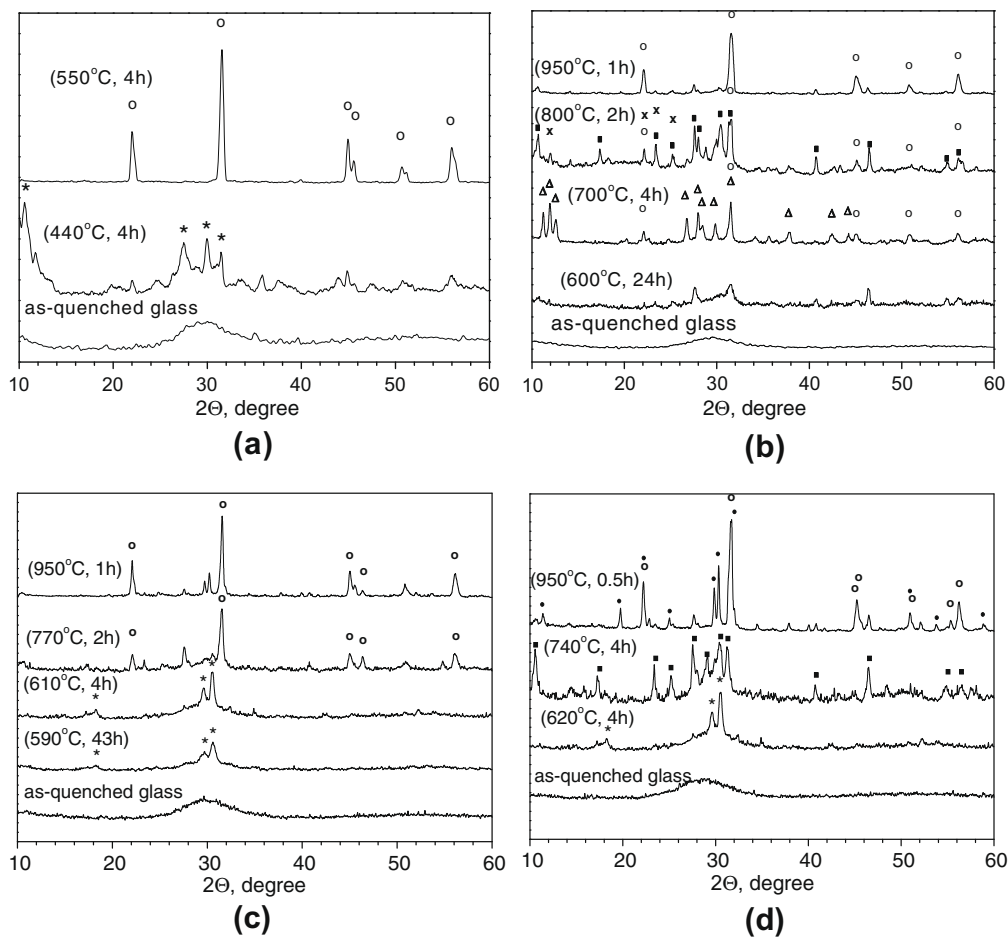


Fig. 2. XRD patterns for glasses heat-treated for various durations at temperatures around exothermic peaks: (a) KNS-0, (b) KNS-5, (c) KNS-15, (d) KNS-20, * – unidentified “N”-phase, x – SiO_2 (JCPDS card #82-1395), ■ – $\text{K}_4\text{Nb}_6\text{O}_{17}$ (JCPDS card #76-977), ○ – KNbO_3 (card #71-0946), Δ – KNbO_3 (JCPDS card #32-0821), ● – $\text{K}_3\text{Nb}_3\text{O}_6\text{Si}_2\text{O}_7$ (JCPDS card #440654).

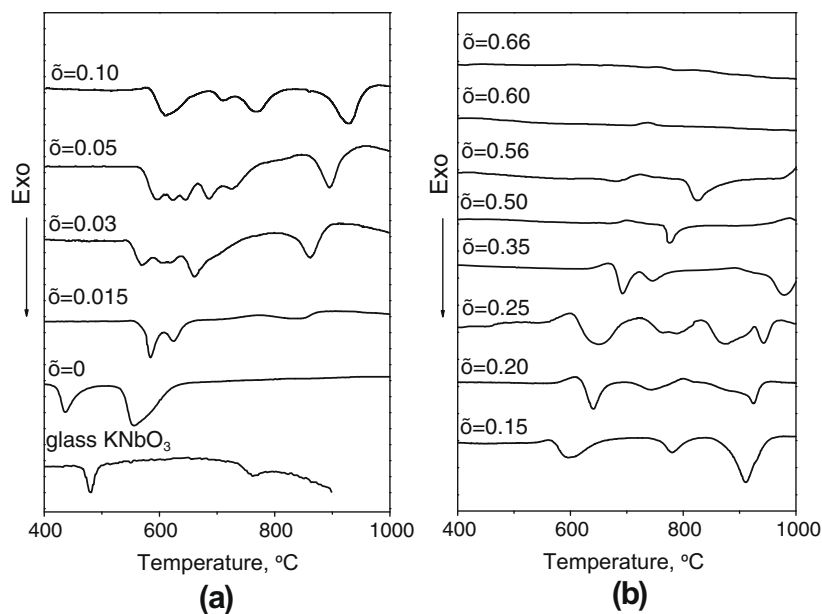


Fig. 3. DTA curves of $(1-x)\text{KNbO}_3 \cdot x\text{SiO}_2$ glass powders of system (a) samples obtained by melt blowing (curve for amorphous KNbO_3 is taken from Ref. [26]), (b) samples obtained by melt pressing (curves for $x = 0.50\text{--}0.66$ are taken from Ref. [7]).

confirmed both for pressed glasses and for air-flow products by the presence of an amorphous halo and the absence of Bragg peaks related to crystalline phases at XRD curves (Fig. 2). Powder samples DTA curves for all as-quenched glasses are shown in Fig. 3. DTA curves for KNS glasses with high silica content as well as for amorphous KNbO_3 [7,27] are also represented.

It can be seen in Fig. 4(a) that the as-quenched KNS-15 glass looks homogenous however a small part of this sample contained inhomogeneities of a few nm in size. The same result was obtained for the KNS-25 glass. Heat treatment at temperatures around the first exothermic DTA peak initiates phase separation, an increase of nanoinhomogeneity and the formation of distinct interphase boundaries (Fig. 4). These changes are evidenced by the formation of two peaks in the range of amorphous halo and the third peak at about 18 degree at XRD curve (Fig. 2(c)). Due to small values of thermal effects and consequent uncertainty of inflection positions an experimental error is relatively high being equal to $\pm 5^\circ\text{C}$ for T_g and $\pm 3^\circ\text{C}$ for T_c .

DTA curves of glass powders obtained by melt spattering with gas flow ($x = 0\text{--}0.1$) as well as the DTA curve of amorphous KNbO_3 (Fig. 3(a)) reveal no inflection related to a glass transition temperature T_g . For these compositions the temperature corresponding to the beginning of the first exothermic peak was taken as T_g by analogy with [27]. Fig. 3(b) shows powder DTA curves for glasses ($x = 0.15\text{--}0.35$) obtained by melt pressing and literature data for glasses with higher SiO_2 content. DTA curves of glasses containing more than 56 mol% SiO_2 do not display exothermic peaks. Compositions, methods of glass production, T_g values and temperatures of maxima of exothermic peaks are summarized in Table 1.

DTA curves of all investigated KNS glasses with SiO_2 content from 0 to 35 mol% (Fig. 3) reveal three pronounced exothermic peaks (for KNS-0 glass there are only two peaks). Typical XRD curves for as-quenched samples and for samples heat-treated at temperatures of exothermic DTA peaks are shown in Fig. 2. They include both glasses obtained by melt spatter with air flow (KNS-0 and KNS-5) and by melt pressing (KNS-15, KNS-20).

Low-temperature exothermic DTA peak for all glasses containing less than 5 mol% SiO_2 is caused by crystallization of unidentified centrosymmetric (no SHG) niobate phase (Fig. 2). At higher SiO_2 content unidentified “N” phase precipitated at temperature of the first exothermic peak. This phase is found to crystallize in glasses with high silica content ($x > 50$ mol%) (Fig. 2(c and d) [7,26].

High-temperature exothermic DTA peak in all cases results from full or partial ferroelectric KNbO_3 crystallization. For KNS glasses with $x \leq 0.2$ ferroelectric KNbO_3 is the only crystallizing phase. The mixture of phases $\text{KNbO}_3 + \text{K}_3\text{Nb}_3\text{O}_6\text{Si}_2\text{O}_7$ (JCPDS card #440654) precipitates in glasses with x equal to 0.25, 0.3 and 0.35. $\text{K}_3\text{Nb}_3\text{O}_6\text{Si}_2\text{O}_7$ content grows with SiO_2 content increase.

There are one or several peaks of various origins in the area between the first and the highest-temperature exothermic peaks in DTA curves. The lack of this exothermic peak in DTA curves of KNS-0 and KNS-1 glasses suggests that it relates to SiO_2 presence in glass composition. The changes in the XRD curves (Fig. 2(b)) confirm this assumption. Below the highest-temperature exothermic DTA peak $\text{K}_4\text{Nb}_6\text{O}_{17}$ (JCPDS card #76-977) and small amount of ferroelectric KNbO_3 are shown to be present along with ceolite-like, cristobalite and α -quartz phases in glasses with SiO_2 content more than 3 mol% (Fig. 2(b–d)). Precipitation of typical phases for low silica content glasses is illustrated in Fig. 4(c). In particular case of KNS-5 fibrous glassy material lines on X-ray powder diagrams (Fig. 5) evidence that after prolonged (10 h or more) heat-treating at temperatures close to the second exothermic peak some SiO_2 -rich phases precipitate in forms of K:ceolite (JCPDS card #82-1395), cristobalite and α -quartz (JCPDS card #82-1395). It is extremely significant that the lines on XRD pattern characteristic for these SiO_2 phases are clearly seen even for crystallized glasses with

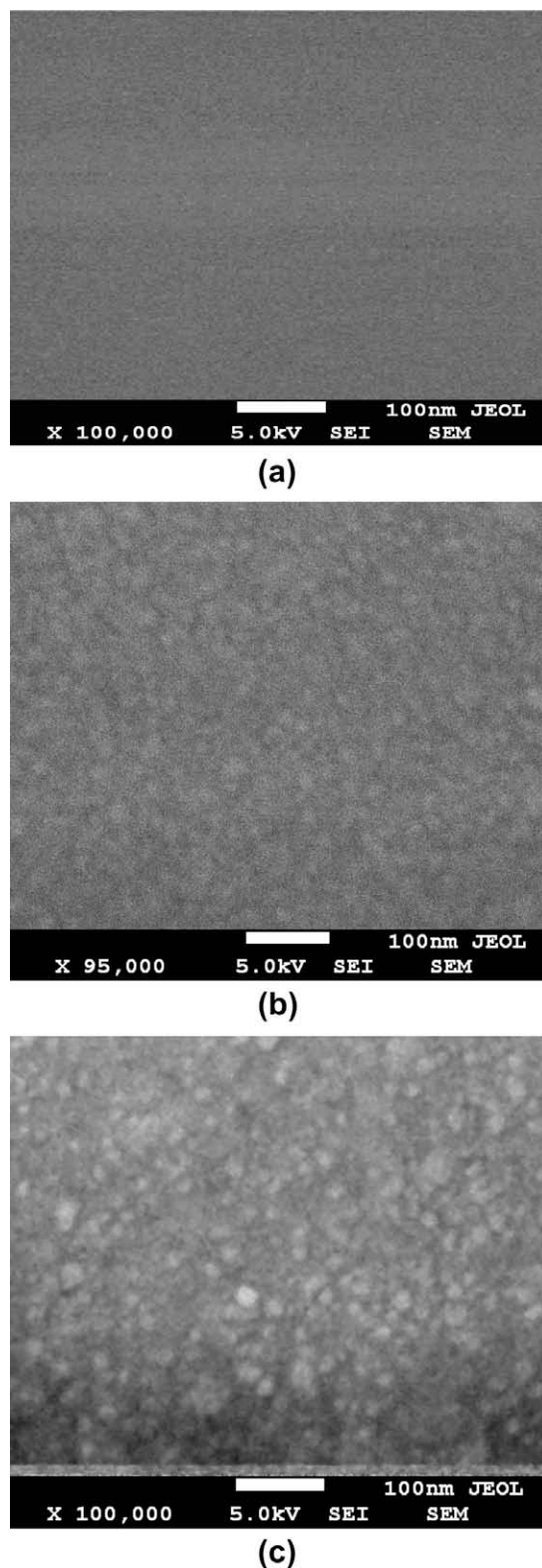


Fig. 4. SEM photographs: (a) as-quenched KNS-15 glass, (b and c) KNS-15 glass heat-treated at 610°C for 2 h (b) and 4 h (c).

SiO_2 content as low as 3 mol%. The only explanation for anomalously strong manifestation of very small amount of silica-based phases may be attributed to their precipitation mainly as a coat on surface of semi-amorphous pieces of niobium-rich fiber-like bodies.

Table 1Compositions of glasses investigated, producing methods, T_g values and temperatures of exothermic peaks maxima at DTA curves, °C (Error estimate: $T_g \pm 5$ °C; $T_c \pm 3$ °C).

Samples	Composition (mol%)			Method	T_g	T_{c1}	T_{c2}	T_{c3} (KNbO ₃)
	K ₂ O	Nb ₂ O ₅	SiO ₂					
KNbO ₃ ^a	50	50	–	TR	455	480	–	760
KNS–0	50	50	–	MB	425	450	–	570
KNS–1.5	49.25	49.25	1.5	MB	545	585	–	620
KNS–3	48.5	48.5	3	MB	550	570	660	865
KNS–5	47.5	47.5	5	MB	570	590	680, 720	890
KNS–10	45	45	10	MB	550	610	710, 765	920
KNS–15	42.5	42.5	15	MP	550	600	782	920
KNS–20	40	40	20	MP	570	640	740	925
KNS–25	37.5	37.5	25	MP	570	650	790, 860	940
KNS–35	32.5	32.5	35	MP	630	690	740	975
KNS–50 ^b	25	25	50	MP	688	775		
KNS–56 ^b	22	22	56	MP	705	825		
KNS–60 ^b	20	20	60	MP	721			
KNS–66 ^b	17	17	66	MP	733			

TR – melt quenching in the twin-roller setup, MB – melt blowing, MP – melt pressing.

^a Data taken from Ref. [23].^b Data taken from Ref. [7].

4. Discussion

A glass-forming region boundary $x = 0.12$ established in the present work is in good agreement with data published by Rao [23] in spite of the fact that their results are obtained for glasses enriched by SiO₂ from the crucible walls during the melting process. This coincidence may be determined by the difference in melt cooling speeds.

Vitreous materials with small addition of SiO₂ are interesting not only for applying thermal poling but also for probable crystallization of ferroelectric KNbO₃. Moreover, doping of pure KNbO₃ by small amount of SiO₂ promises easier obtaining macroscopic samples $(1 - x)\text{KNbO}_3 \cdot x\text{SiO}_2$ ($x \sim 0.01\text{--}0.1$) comparative to synthesis of pure KNbO₃ via a technique [27] based on rolling the melt of the same composition between twin metallic rollers and hydrostatic pressing at high pressure in order to consolidate them into continuous solid material.

Data given in Fig. 3 and Table 1 allow drawing T_g vs. glass-former content c dependence in the full range from 0% to 100% (Fig. 6). 1250 °C is taken for silica glass (i.e., $c = 100\%$) T_g value.

$T_g(c)$ dependence is complicated and non-monotonic. One can distinguish three characteristic sections at this curve. Small addition of SiO₂ substantially enhances glass-forming ability of potassium niobate melt resulting in abrupt growth of T_g by 120 °C between KNbO₃ and KNS-1.5 (i.e., 1.5 mol% SiO₂). Further addition of SiO₂ (3, 5, 10 mol% and up to about 25 mol%) virtually does not influence on T_g value keeping it at the level close to one for 1.5 mol% SiO₂.

Such a response of T_g level to adding the glass-former into the KNbO₃ melt resembles a reverse response of silica to adding a small amount of alkali oxides. According to [28], even a rather small addition of 0.32 mol% Na₂O dramatically increases crystallization ability of silica glass but further additions of Na₂O up to 1.36 mol% cause merely slight crystallization ability growth (Fig. 6(b)).

In the range from 1.5 to ~15 mol% SiO₂, T_g is nearly constant and in the range of 15–66 mol% SiO₂ increases almost linearly, being slightly convex upwards centered around 50 mol%. If extrapolated to higher SiO₂ content this linear dependence will give T_g less than 800 °C for silica glass. That is 450 °C less than real silica glass T_g value. It means that in the range 70–100 mol% SiO₂ T_g grows substantially faster with increase of SiO₂ content. A nominal extrapolation of $T_g(c)$ to real T_g value of silica glass taking into account numerous data about strong dependence of silica glass viscosity on impurities concentration [29] is shown in Fig. 6(a).

The features mentioned show possibility of non-monotonic changes of KNS glass structure (in particular changes of Nb atoms coordination), crystallization behavior and glassy-state immiscibility processes due to x growth. KNS glasses with 50–60 mol% SiO₂ occur in the metastable immiscibility region [7,30,31]. Prolonged heat-treatment of KNS-50, KNS-56 and KNS-60 glasses at temperatures slightly higher than T_g results in their opalescence. This opalescence is quite strong in KNS-50 glass and barely noticeable in KNS-60 glass whereas KNS-66 glass did not reveal signs of opalescence at all [7,30]. It is obvious that KNS compositions from 66 mol% to pure SiO₂ lie outside the liquid–liquid immiscibility region. As for glasses containing less than 50 mol%, one can assume that they are also inclined to phase separation as well as KNS-50, KNS-56 and KNS-60 glasses. T_g growth along with x increase in the range 20–66 mol% SiO₂ no seems to be slower comparative to T_g growth in the range 70–100 mol% SiO₂ due to phase separation possibility in the wide range below 66 mol% SiO₂. Meanwhile abnormal character of $T_g(c)$ in the range 5–20 mol% SiO₂ suggests one more mechanism of phase separation.

As well as it was shown by Sato et al. [27] DTA curve for amorphous KNbO₃ contains two exothermic peaks (Fig. 3). By means of X-ray diffraction it is shown that the first one corresponds to crystallization of centrosymmetric KNbO₃ phase (JCPDS card #32-0821) together with another metastable unidentified phase while the second peak corresponds to crystallization of ferroelectric KNbO₃ (JCPDS card #71-0946) (Fig. 2(a)). It should be mentioned that characteristic points at DTA curve of KNS-0 sample (Fig. 3) are substantially shifted to lower temperatures in comparison with those of KNbO₃ given in [27]: T_g , T_{c1} and T_{c2} are shifted by 30, 30 and 190 °C, respectively.

Another feature deserving notice is a variety of crystalline phases precipitating in the pure KNbO₃ samples heat-treated around the first exothermic DTA peak: Sato et al. [27] reported crystallization of a single centrosymmetric KNbO₃ phase (JCPDS card #32-0821), whereas in KNS-0 glass obtained in the present study an unidentified crystalline phase precipitated mainly together with small amounts of centrosymmetric KNbO₃ (JCPDS card #32-0821) and even the slight addition of ferroelectric KNbO₃ (JCPDS card #71-0946). Remarkable difference in crystallization behavior must be related to different thermal properties of the samples. This difference can be seen very clearly at the position of the second exothermic DTA peak ($T_{\text{max}2} = 570$ °C in the present work, $T_{\text{max}2} = 760$ °C [27]). It is evidently due to large number of crystallization centers in glass obtained in the present work since it was obtained in more non-equilibrium state and different

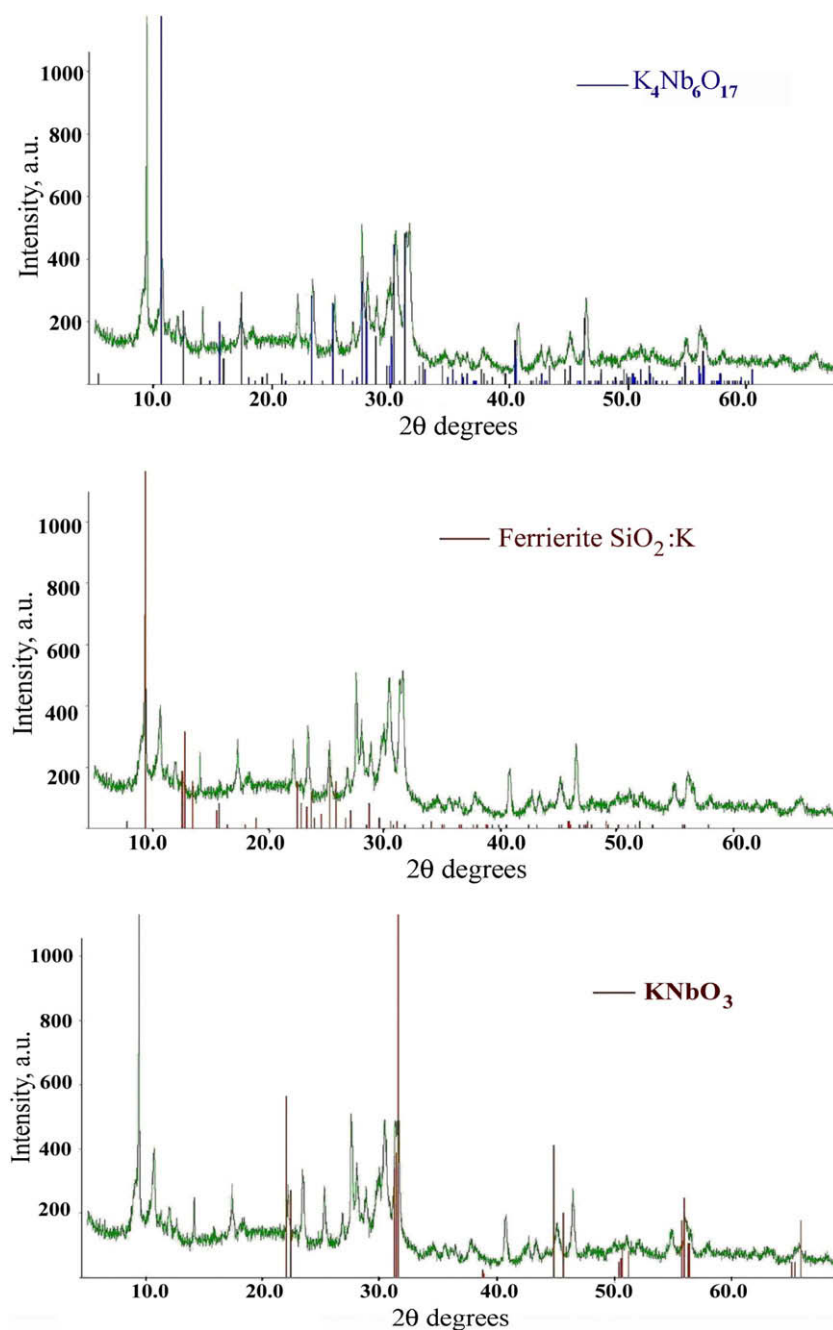


Fig. 5. X-ray patterns of $\text{KNbO}_3\text{:}5\text{SiO}_2$ glass heat-treated for 24 h at 600 °C with indications of lines corresponding to crystalline phases denoted in this figure.

regions of the melt stream were cooled down at different speeds. Another possible reason for a large difference in KNbO_3 precipitation from KNS-0 glass can be the fact that KNS-0 composition was slightly modified to compensate K_2O evaporation from the melt. This measure provided maximal proximity to exact KNbO_3 composition while nothing no K_2O evaporation compensation is reported by authors of the work [27].

DTA curves of KNS glasses with low SiO_2 content (Fig. 3) contain several exothermic peaks. X-ray diffraction of the samples heat-treated in the temperature range from T_{c1} to T_{c3} (Fig. 2), allows conclusion that lower-temperature effect with the maximum at T_{c1} is related to amorphous phase separation and corresponding precipitation of unidentified centrosymmetric crystalline phase. Formation of transparent glass-ceramics is revealed in the composition range $x = 0.15\text{--}0.35$ (Fig. 7). It is

confirmed by X-ray diffraction analysis (Fig. 2(c and d)) and electronic microscopy (Fig. 4(a–c)). It should be noted that inhomogeneities almost do not grow in the wide temperature range. For instance 4 h long heat-treatment at 610 °C which is 60 °C higher than T_g for KNS-15 glass inhomogeneities increase barely up to 20–30 nm. However, interphase boundaries become much more pronounced for longer heat-treatment. This suggests that in KNS glasses studied a chemical differentiation and structural ordering processes take place at temperatures around T_g keeping transparency and without noticeable increase of inhomogeneities size. As this process can be controlled by heat-treatment conditions variation and concentration gradient near the interphase boundaries can be gradually changed one can assume according to approach developed in [7] that the revealed inhomogeneities appear to account for SHG origin.

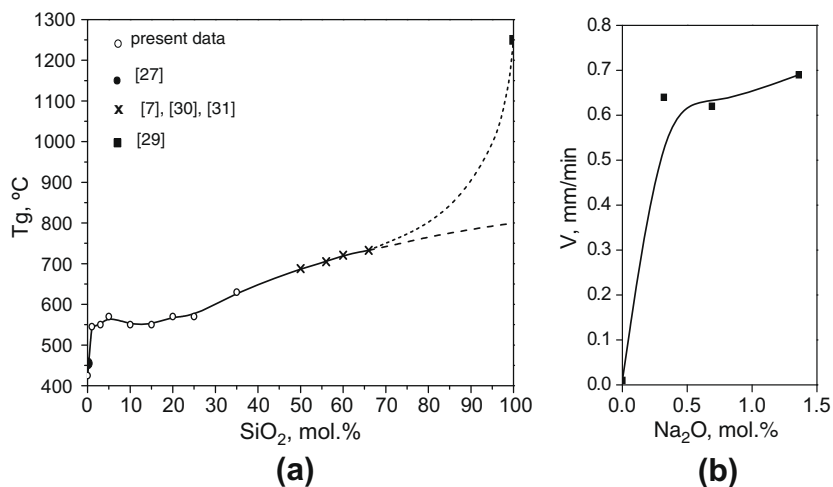


Fig. 6. (a) T_g vs. SiO_2 content for KNS glasses ($\text{K}_2\text{O}/\text{Nb}_2\text{O}_5 = 1$), errors for data of the present work lie within circles. (b) Crystallization rate vs. Na_2O content for precipitation of silica and three sodium-silicate phases at 1450 °C (data are taken from Ref. [29]). Both lines are drawn as guides to the eyes.



Fig. 7. Photograph of KNS-15 glass sample heat-treated at 610 °C during 4 h.

For glasses with $x = 0.03$ – 0.1 the second exothermic DTA peak corresponds to SiO_2 precipitation on the surface of $\text{K}_4\text{Nb}_6\text{O}_{17}$ -rich semi-amorphous flakes or fibers. It was also confirmed for the glass with $x = 0.05$ by XRD patterns (Fig. 2(b)). Crystallization of the glasses containing less than 15 mol% SiO_2 around the second exothermic DTA peak leads to formation of centrosymmetric crystalline phases mixture with slight addition of ferroelectric KNbO_3 (JCPDS card #71-0946) whose concentration grows as the heat-treatment temperature and duration increase. A crystallization product of glasses with $x = 0.15$ – 0.35 is $\text{K}_4\text{Nb}_6\text{O}_{17}$ (JCPDS card #76-977), which at higher heat-treatment temperatures and durations turns into ferroelectric KNbO_3 (JCPDS card #71-0946) according to XRD (Fig. 2) and SHG results (Fig. 8).

The DTA peak at the highest temperature should be therefore assigned to ferroelectric potassium niobate crystallization for glasses with $x = 0$ – 0.2 as XRD data in Fig. 2 indicate. Temperature dependence of SHG signal for KNS glasses crystallized near the third exothermic DTA peak illustrates this conclusion. It is also corroborated with heat-treated KNS glasses SHG values: after heat treatment at temperatures higher than 900 °C these values become several thousand times as large as SHG of the powder quartz standard, clearly witnessing presence of structurally perfect KNbO_3 polycrystalline material. Anomalies in SHG dependence on temperature seen in Fig. 8 for KNS-0 and KNS-5 glasses coincide with structural phase transitions known to take place in chemically pure ferroelectric KNbO_3 crystals at 220 and 430 °C. Distinct thermal

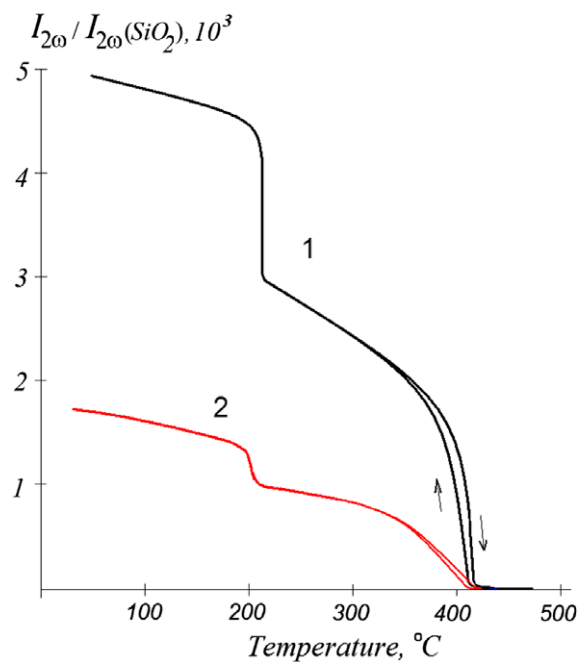


Fig. 8. SHG intensity relative to $\alpha\text{-SiO}_2$ powder standard vs. temperature for KNS-0 (curve 1) and KNS-5 (curve 2) glasses crystallized near the third exothermic DTA peak. Arrows show temperature change direction in the heating-cooling cycle.

hysteresis in heating-cooling cycle at around 430 °C corresponds to first-order ferroelectric phase transition. Absence of SHG signal above this transition agrees with centrosymmetric crystal structure of KNbO_3 in its paraelectric phase. It is demonstrative that addition of SiO_2 to glass composition strongly affects SHG intensity from crystallized powder and at the same time changes phase transitions parameters very slightly or does not change them at all. This confirms the picture of SiO_2 coating of KNbO_3 grains in crystallized materials. Judging by diminishing of SHG signal from different forms of spattered KNS glasses this coating is especially consistent in the case of thin fibers obtained from spattered melt.

Glasses obtained as plates are amorphous according to XRD data and contain a significant quantity of niobium-oxygen polyhedra which is very promising for obtaining materials with high second-order optical non-linearity by thermal poling. Moreover,

thermal poling applied to nanostructured glasses is expected to be much more effective comparative to homogeneous glasses. Applying such materials in photonics can be very fruitful via technique described in [27] it is possible to produce thin glass plates containing 1–10 mol% SiO₂ which are evidently even more promising for thermal poling.

5. Conclusion

Glass-forming and crystallization properties of $(1-x)\text{KNbO}_3 \cdot x\text{SiO}_2$ glasses with x varying from 0% to 35% are studied. It is shown that in these glasses a crystalline $\text{K}_4\text{Nb}_6\text{O}_{17}$ phase and a small amount of ferroelectric KNbO_3 along with ceolite-like, cristobalite and α -quartz SiO_2 phases precipitate at low temperatures, whereas ferroelectric KNbO_3 precipitates at temperatures above 800 °C. Ferroelectric crystals which precipitate in the bulk glass are shown to produce a high SHG signal. Thin transparent plates can be obtained by quenching via pressing between steel plates for glasses with $x \geq 12\%$. Transparent glass-ceramics can be obtained by heat-treatment near the first exothermic DTA peak in a wide range of glass compositions. Low-silica KNS glasses and fine transparent glass-ceramics allowing a remarkable SHG can be formed by means of the specified heat-treatment.

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